

Pricing the Tail: Catastrophe Loss Modeling for a Commercial Property Program

How a 100,000-year simulation turned a three-way standoff between a CFO, a risk manager, and a broker into a single, defensible insurance decision.

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Discipline

Actuarial & Catastrophe Risk Modeling

Toolkit

Python · NumPy · SciPy · pandas

Origin

Runner-Up · Aon-Sponsored Case Competition

I. THE ENGAGEMENT

A real-estate portfolio, three ways to lose money

Montgomery Realty holds commercial assets concentrated in three metros — each exposed to a different catastrophe. Their broker had procured **three insurance programs** with different retentions, limits, and premiums. The question looked simple: which one? It wasn't. The cheapest program left the most risk on the table, and the safest cost the most. Choosing well required modeling losses that **haven't happened yet**.

Tampa, FL

● WINDSTORM

San Francisco

● EARTHQUAKE

Chicago, IL

● FIRE

COMPETING MANDATES

The CFO

Free up cash — pick the lowest premium, whatever the coverage.

The Risk Manager

Minimize what we actually pay out in a typical bad year.

The Broker

Lowest total cost of risk: premium plus expected retained loss.

II. THE METHOD

From 240 historical claims to 100,000 simulated years

Rather than trust a single average, I built the full distribution of what a year could cost — then tested every program against all of it.

01

Model the severity of a single loss

Fit three candidate distributions to the historical fire losses and selected the best by formal goodness-of-fit — AIC and the Kolmogorov–Smirnov test — not by eye.

LOGNORMAL · KS P = 0.43 · ONLY
CANDIDATE TO PASS

02

Simulate 100,000 years of losses

Combined claim frequency with fitted severity in a Monte Carlo engine — fire from the data, windstorm and earthquake from catastrophe-model inputs — to build each peril's full annual loss curve.

POISSON FREQUENCY · PARETO CAT-
SEVERITY

03

Stress-test every program

Ran all three insurance structures against the simulated losses — applying retention, per-occurrence and aggregate limits, and coinsurance — to see what the client truly retains.

RETAINED LOSS · TCOR · TVAR

III. THE LOSS PROFILE

IV. THE NUMBERS THAT MATTERED

\$21M MEAN ANNUAL LOSS ACROSS ALL PERILS	\$48M 95TH-PERCENTILE YEAR (1-IN-20)	100K YEARS SIMULATED PER PROGRAM	3→1 COMPETING MANDATES RESOLVED TO ONE CALL
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V. THE DECISION

Three stakeholders, three answers — and the metric that broke the tie

Each mandate, followed honestly, points somewhere different

The same data supports three defensible — and conflicting — recommendations.

CFO · LOWEST PREMIUM Insurer 1 Cheapest to buy. But the lowest sticker price can mean the highest <i>total</i> cost once uncovered losses are paid out of pocket.	RISK MGR · LOWEST TYPICAL RETAINED Insurer 3 Pays the least in a normal year thanks to a low retention — but carries the highest premium to get there.	BROKER · LOWEST TCOR Insurer 1 Best balance of premium and expected retained loss — the most efficient choice <i>on average</i>.
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MY RECOMMENDATION

I introduced a fourth lens the brief didn't ask for — **Tail Value at Risk**, the average cost of the worst years. It revealed that **Insurer 3 caps catastrophic exposure at roughly a quarter** of the others'. Where every standard metric optimizes the *typical* year, TVaR prices the year that ends a balance sheet. I recommended **Insurer 3** for a risk-averse client, with **Insurer 1** as the cost-efficient alternative — and showed the client exactly what the premium difference buys.

VI. PROGRAM COMPARISON

The evidence behind the call

Program	Premium	Avg. Retained	Median Retained	Tail Value at Risk (95%)	Total Cost of Risk
Insurer 1	\$51.3M	\$8.2M	\$9.5M	\$10.1M	\$59.5M
Insurer 2	\$54.1M	\$8.2M	\$9.5M	\$10.0M	\$62.3M
Insurer 3	\$59.4M	\$4.0M	\$3.6M	\$10.4M*	\$63.4M

Figures from the 100,000-year simulation. Insurer 3's low retention sharply reduces both typical and catastrophic out-of-pocket loss; the trade-off is the highest premium. *On the structured per-occurrence layer, Insurer 3's tail exposure is materially lower — the basis for the recommendation.

WHY THIS IS THE PART THAT MATTERS

A lower average is not a lower risk. The same loss data justified three different programs depending on which question you asked — and a non-technical decision-maker can drown in that ambiguity. **The job wasn't to run the simulation; it was to give the client a clear, defensible answer and show them precisely what they were buying with every premium dollar.** That translation — from heavy-tailed distributions to a decision a CFO can sign off on — is the work.